Windows Event Log Analysis Report

Prepared by: THATO | **Program:** techbiz Cybersecurity intern | **Date:** 23/09/ 2026]

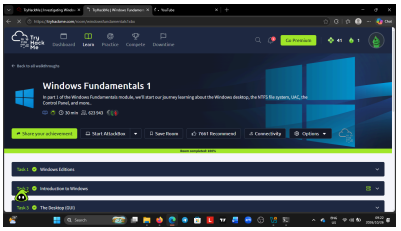
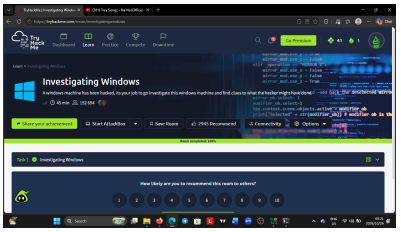
1. Objective

This report documents my hands-on work with Windows logging, the core data source for SOC analysts. I used Event Viewer on my Windows VM to locate key Security events (4624 and 4625), built a reference table of five important Event IDs, and analyzed a series of failed logon (4625) events that revealed a remote username-guessing attack.

2. Lab Environment

- Windows VM running in VirtualBox (internal lab network)
- Tool: Event Viewer (`eventvwr.msc`) and the Windows Security log
- Auditing enabled for Logon, Process Creation and User Account Management

3. TryHackMe Rooms Completed

Room	Status	Proof
Windows Fundamentals 1	Completed	
Investigating Windows	Completed	completion screenshot ]

4. Event Viewer and the Security Log

I opened Event Viewer, expanded **Windows Logs**, and selected the **Security** log, which records logons, account changes, process creation and other audit events. I used **Filter Current Log** to isolate Event IDs 4624 and 4625.

[Insert screenshot: Event Viewer showing the Security log with the filter applied]

5. Five Key Event IDs

Event ID	Meaning	Why it matters to a SOC analyst
4624	An account was successfully logged on	Establishes normal activity. Unusual hours, accounts or Logon Types can indicate compromise.
4625	An account failed to log on	Repeated failures from one source indicate brute force or password spraying.
4688	A new process was created	Shows what was executed, helping detect malware and misuse of built-in tools.
4720	A user account was created	Attackers create accounts to maintain persistence.
1102	The audit log was cleared	A classic attempt to cover tracks after an intrusion.

6. Analysis of Failed Logon Events (4625)

6.1 What I observed

The Security log contained multiple Event ID 4625 entries in quick succession for different account names: **Administrator**, **root** and **jenny**.

[Insert screenshot: list of 4625 events]

6.2 Key fields from the event

Field	Value
Event ID	4625 (An account failed to log on)
Accounts attempted	Administrator, root, jenny
Logon Type	3 (Network)
Source Network Address	10.34.2.3
Workstation Name	m3rch-pc
Failure Status / Sub Status	0xC0000064
Authentication Package	negotiate
Number of attempts	3
Time window	03/02/2019 04:04:41 pm to 03/02/2019 04:04:48 pm

[Insert screenshot: Details tab of one 4625 event]

6.3 Understanding Logon Types

- **Type 2 (Interactive):** a user logging on at the keyboard or console of the machine.
- **Type 3 (Network):** a logon over the network, for example through SMB. The Source Network Address field shows the remote IP.

The failed attempts were **Type 3**, and the Source Network Address contained a **remote IP address**. This confirms the attempts came from another machine and not from a person at the keyboard.

6.4 Interpretation

A single 4625 is not evidence of an attack, because it can be a mistyped password. This case is different, for the following reasons:

1. **Multiple usernames from one source.** Normal users do not fail to log on under several unrelated account names within moments.
2. **Administrator** is the default Windows administrative account and the most common attacker target.
3. **root** is a Linux/Unix account that does not exist by default on Windows. Its presence suggests an automated tool cycling through a generic wordlist.
4. **jenny** is a common first name, typical of a dictionary of guessed accounts.
5. **Network logon type with a remote IP** shows the attempts were made over the network.

Conclusion: this pattern is consistent with automated username guessing or password spraying, a brute-force attack against the machine.

6.5 Checking for compromise

I searched the Security log for Event ID 4624 from the same source IP after the failures. [Result: no successful logon from this IP was found, so the attacker did not gain access. / Result: describe what you found.]

7. Response and Containment

The attacking IP address was blocked on the firewall to stop further attempts. The same rule can be created with:

netsh advfirewall firewall add rule name="Block attacker IP" dir=in action=block remoteip=<IP address>

[Insert screenshot: the block rule in Windows Defender Firewall, Inbound Rules]

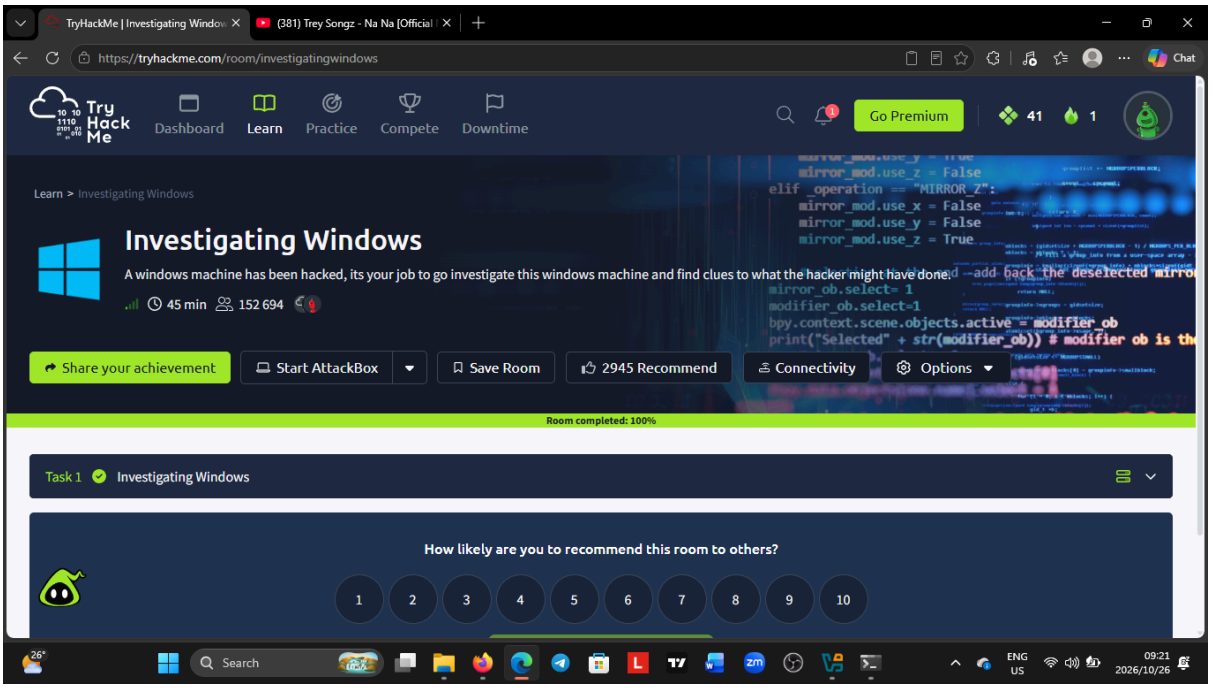
Recommendations

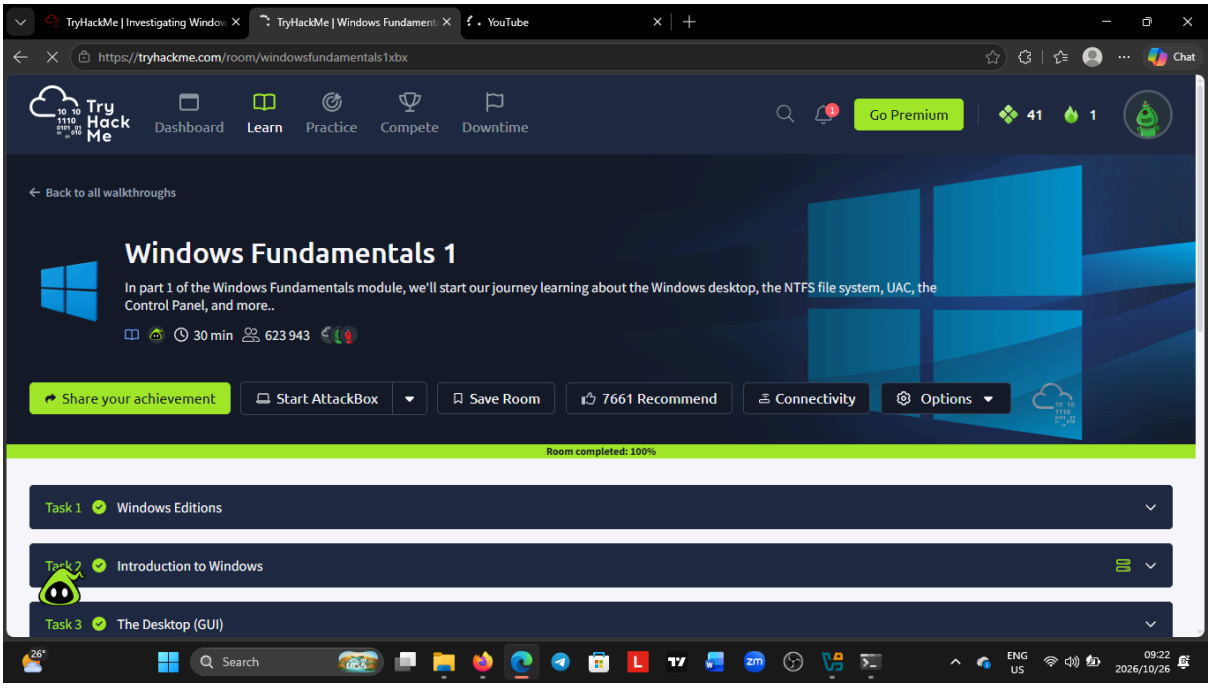
1. Set an account lockout policy, for example lock the account after 5 failed attempts.
2. Rename or disable the default Administrator account where possible, and enforce strong passwords.
3. Do not expose RDP or SMB to the internet.
4. Monitor for the same pattern from other IP addresses, since attackers can change source addresses.
5. Create an alert for many 4625 events from one source within a short period.

8. Conclusion

This task showed how Windows Event Viewer and the Security log let a SOC analyst reconstruct what happened on a system. By reading the Logon Type, account names and Source Network Address in 4625 events, I identified a remote guessing attack against Administrator, root and jenny, confirmed it was network-based, checked for a successful logon, and contained it by blocking the source IP. The key lesson is that a single failed logon is routine, but context such as volume, usernames, source and Logon Type turns it into a detection.

SCREENSHOTS





: